2.2 Preparation and training methods in relation to maintaining and improving physical activity and performance	2.2.1	Knowledge and understanding of preparation and training methods in relation to maintaining and improving physical activity and performance.
	2.2.2	Fitness tests: functional thresholds, lactate threshold/anaerobic threshold/maximum steady state, gas analysis, multi-stage fitness test, step tests, yo-yo test, Cooper minute run, Wingate test, maximum accumulated oxygen deficit (MAOD), RAST (repeat anaerobic sprint test), Cunningham and Faulkner, jump tests, Margaria-Kalaman, strength tests, agility tests, sprint tests < 100 m.
	2.2.3	Interpret, calculate and present data (tables and graphs) based on fitness test results.
	2.2.4	Determinants of movement/running performance and their application to sprint, endurance and intermittent activities.

Subject content	What students need to learn
	2.2.5 Components of fitness: localised muscular endurance, $\dot{V}O_2$ max, anaerobic capacity, maximal strength, strength, power, speed, agility, coordination, reaction time, balance, flexibility, exercise economy, maximal and 'submaximal' aerobic fitness.
	2.2.6 Principles of training: individual needs, specificity, progressive overload, Frequency Intensity Time and Type (FITT), overtraining, reversibility.
	2.2.7 Different ways of measuring and calculating intensity: percentage of one repetition maximum (RM), Rate of Perceived Exertion (RPE), percentage of functional threshold, target HR, work to rest ratios.
	2.2.8 Target heart rate: understanding and use of Karvonen's theory.
	2.2.9 Contemporary technologies used by the performer and coach to monitor fitness and performance.
	2.2.10 Periodisation: Macro, Meso and Micro Cycles Knowledge and understanding of the preparation phase (general and specific), competition phase and transition phase.
	2.2.11 Methods of training and their appropriateness for different activities: interval, circuits, cross, continuous, fartlek, flexibility (static, ballistic and proprioceptive neuromuscular facilitation (PNF)), weights (free weights and machines), resistance (including pulleys, parachutes), assisted (including bungees, downhill), plyometrics, speed agility quickness (SAQ) and functional stability. Advantages and disadvantages of each method of training.
	2.2.12 Preparation for performance at altitude, in heat and in humidity.
	2.2.13 Knowledge and understanding of strategies for speeding up recovery following physical activity: cooling down, massage, ice baths, compression clothing.